Installation instructions (to be completed before the workshop)

For this workshop, you will need:

- **Mac M1 or higher.**
- **Android Studio Quail 1 | 2026.1.1 Patch 2**.
- **JDK 21 installed & configured -** note new Panda feature to setup the Gradle JVM (I have Amazon Cornetto 21). Note for Desktop Hot Reload you need to use a JetBrains JVM.
- Cloned repository: https://github.com/pahill/deep-dive-into-kmp.
- **Xcode 26.5 (for macOS)** installed and configured with simulators.
- I have tested with Android emulator / iOS simulator and device / desktop (JVM) on MacOS. I have not been able to test on an Android device, so don't panic if it doesn't work - just use an emulator.

If there are newer versions of Android Studio or Xcode, you can attempt to use them, but I haven't tested them yet. I would strongly recommend staying with the versions I stated above, please.

Please see sections below for details.

Problems/Questions?

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Android Studio setup

The instructions below describe the minimum setup to be able to run Android/Desktop. Please follow these instructions for macOS, Windows or Linux.

Pre-steps/knowledge:

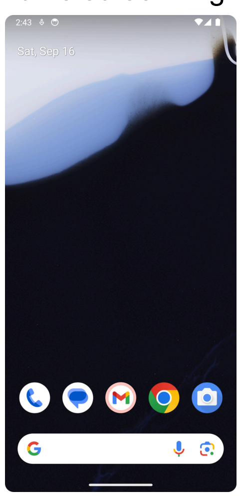
1. You will need to install the latest stable Android Studio.
2. You will need to install the JDK 21 or higher**.**
3. If you're not sure whether your Mac has an Intel or Apple Silicon processor, follow the guide here.

Instructions:

1. Navigate to https://kotlinlang.org/docs/multiplatform/quickstart.html and follow the installation instructions.
2. In Android Studio, get the workshop repository from GitHub.
    a. Click on the "Get from VCS" button (assuming no other projects are open).
    b. In the "Get from Version Control" screen:
        i. Set "URL" to: https://github.com/pahill/deep-dive-into-kmp.git

ii. Set “Directory” to where you want the project to be saved (and take note where you’re saving to).
iii. Click on the “Clone” button.
iv. Note this may take some time.

4. In Android Studio, set up an emulator with API 24 or above. Assuming you have a project open:
   a. Navigate to “Tools” -> “Device Manager”
   b. In the “Device Manager” panel, click on “+” for adding a new device.
   c. In the dropdown that appears, select “Create Virtual Device”.
   d. In the dialog, select “Phone” for Category, and any phone that has the Play Store installed (the arrow icon ). I used Pixel 8. Click the “Next” button.
   e. In the “Select system image” panel, click on the download button(s)  next to the release name at the top of the list. The “SDK Component Installer” dialog box will appear and install the correct system image. I used “CinnamonBun” for API 37 (not the Preview). This may take some time to download and install. Click the “Finish” button.
   f. Back in the “Configure virtual device” screen, give your emulator a name in the “Name” box, and click the “Finish” button.
   g. To verify that you have set up the emulator correctly:
      i. Go to “Tools” -> “Device Manager” again.
      ii. Click the launch button () corresponding to the emulator you created.
      iii. The device will then boot up, and you’ll be taken to the home screen. Your home screen might look something like this:



iOS setup

After you have installed and set up Android studio as described above, follow the instructions below to make sure iOS components are set up properly. Note: this is for macOS users only.

Pre-steps/knowledge:

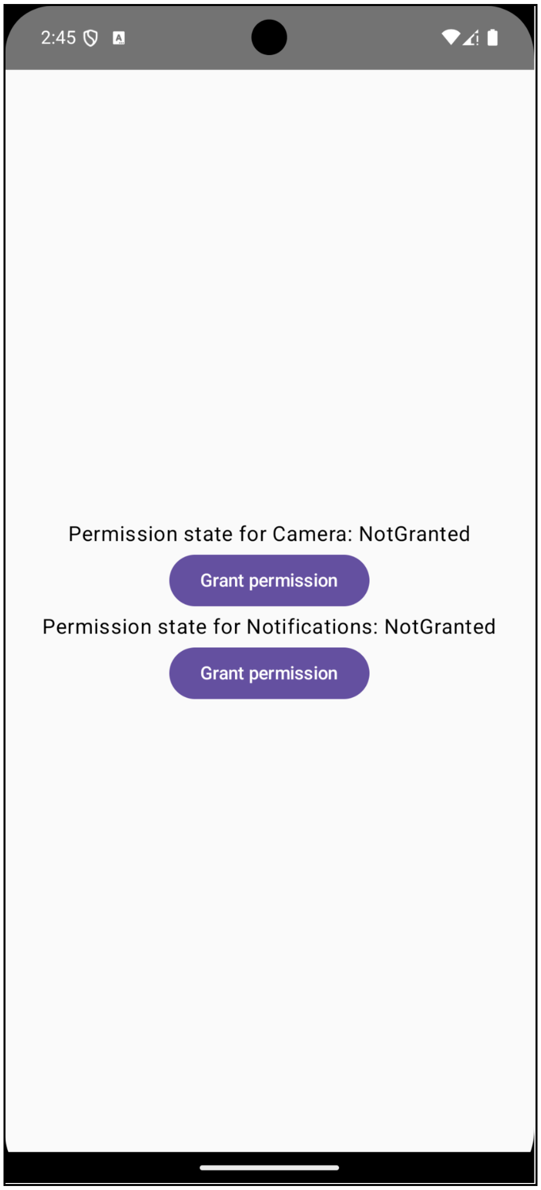
1. Select the default devices when prompted by Xcode that you want to develop for, like iOS and macOS.

Installation steps:

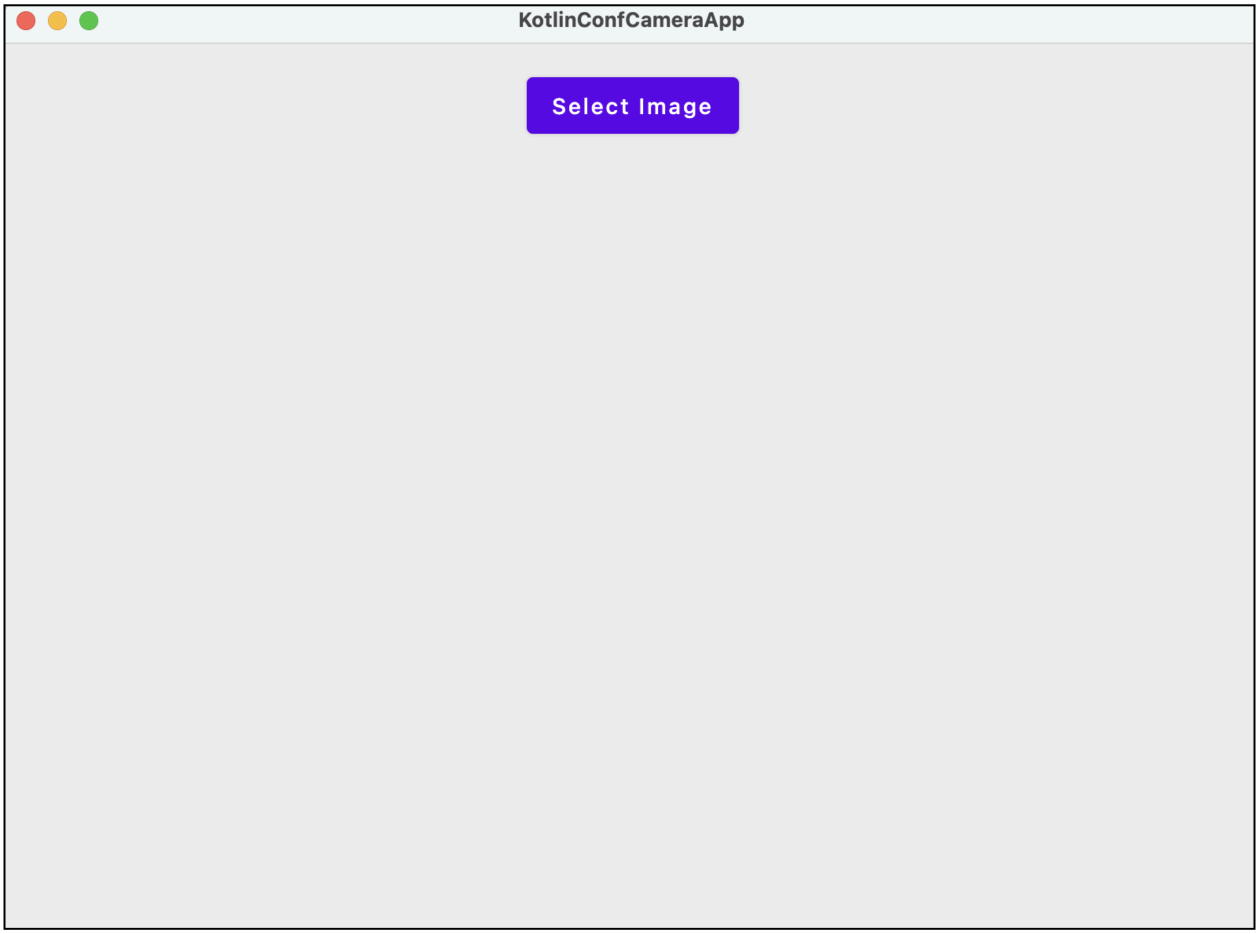
1. Navigate to https://kotlinlang.org/docs/multiplatform/quickstart.html and follow the installation instructions, also follow the Xcode instructions. Make sure all the Preflight checks have passed.
2. Set up your phone (if you have one / want to):
    a. Enable developer mode on your iPhone by following these instructions.
    b. Set up Xcode:
        i. Add your Apple Account to the Apple Accounts settings in Xcode:
        ii. Choose a valid team in your project's Signing & Capabilities pane:
            1. In the right-hand pane, click on "iosApp"
            2. Then select "iosApp" under Targets.
            3. Click on the "Signing & Capabilities" button to open the pane.
            4. Select your account under Team. In my case it says Pamela Hill (Personal Team).
    c. To pair a device with a physical connection:
        i. Connect the device to your Mac using **an appropriate cable**.
        ii. Unlock the device and follow any instructions that appear in Xcode or on the device to trust the computer/device.

Workspace verification

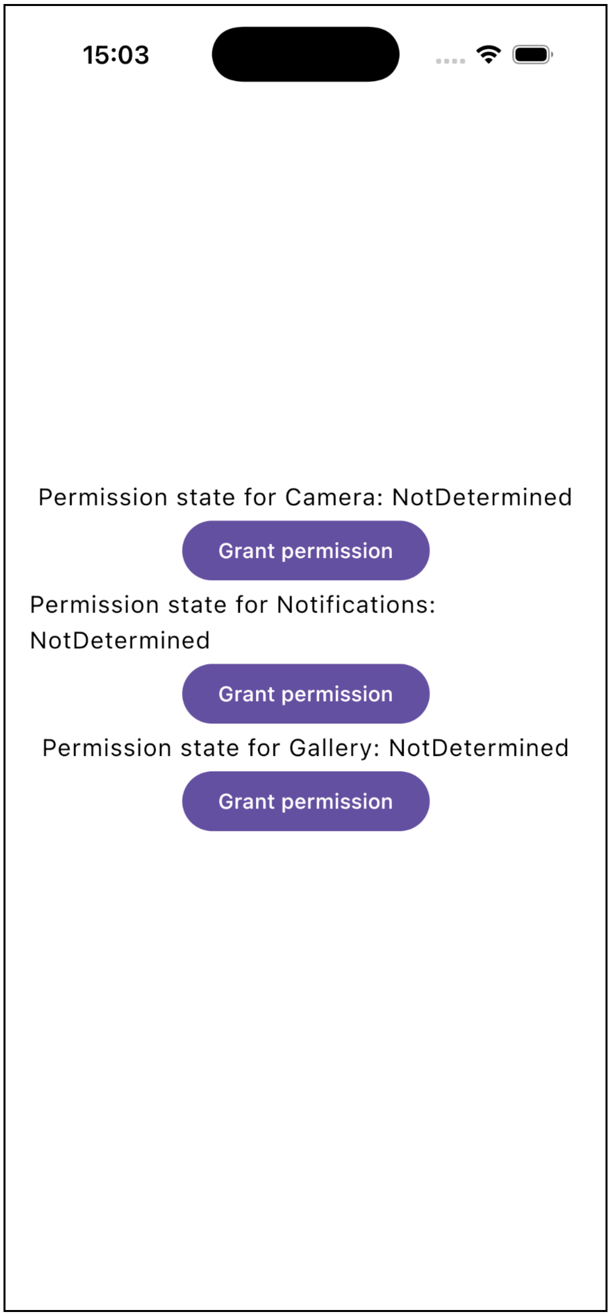
1. In Android Studio, navigate to "File" -> "Open". In the file dialog, navigate to your local repository and then to the subdirectory /full-example. The project should be compilable and runnable, but may need some time for Gradle to finish syncing.
2. To run the project on an Android emulator, ensure that "androidApp" is selected in the "Run configurations" dropdown, that the emulator you created in the installation steps is selected, and click on the "  " button (to the right). The app should start and the following screen should appear.



3. To run the project on the desktop, ensure that “desktopApp [jvm]” is selected in the “Run configurations” dropdown, and click on the “  ” button (to the right). The app should start and the following screen should appear.



4. **Close the desktop application before proceeding to the next step. There seems to be a bug in Android Studio that requires this action.**
5. To run the project on an iPhone, ensure that “iosApp” is selected in the “Run configurations” dropdown, and click on the “  ” button (to the right). The app should start and the following screen should appear.



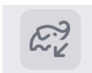
Troubleshooting:

- If you are using a physical iPhone, you will need to make sure you are signing the app with your account.
    1. Setting the team in the Xcode project:
        a. In the right-hand pane, click on “iosApp”.
        b. Then select “iosApp” under Targets.
        c. Click on the “Signing & Capabilities” button to open the pane.
        d. Select your account under Team. In my case it says Pamela Hill (Personal Team).

- Make sure your physical device is unlocked, and listed as Execution Target in Android Studio.
- There might be a bundle identifier error, simply edit the identifier to make it unique.

Syncing all projects

To save ourselves a lot of time, let's download our dependencies beforehand. For each project listed here, open the project and sync with Gradle using the icon top-right in the toolbar, it looks like a little elephant



- full-example
- practical1/starter
- practical1/final
- practical2/starter
- practical3/starter
- practical3/final
- practical4/starter

Optionally, you can also sync:

- kover
- lldb_example
- xcode_kotlin_example